TASK 4

Data Storytelling & Statistical Validation

Business Narrative • Presentation Outline • A/B Test Example

Data note
This document demonstrates the method using hypothetical example data. The figures are not measured results from ApexPlanet or a real company. Replace them with results from your own dataset before final submission.

1. Executive Story

Business question: Did a redesigned checkout improve the website conversion rate? The story should move from the business problem, to evidence, to statistical validation, and finally to a decision. A change in observed conversion alone is not enough; uncertainty and practical impact must also be considered.

2. Example A/B Test

Group	Visitors	Orders	Conversion
Control (A)	1,000	100	10.0%
Variant (B)	1,000	120	12.0%

Observed difference: +2.0 percentage points; relative lift: +20%. These descriptive results are promising, but a statistical test is required before claiming the change is reliable.

3. Hypothesis & Test Selection

Null hypothesis (H■): The conversion rates are equal in the two groups.
Alternative hypothesis (H■): The conversion rates differ.
Test: Two-proportion z-test (or a chi-square test for a 2×2 table). Use a two-sided test at $\alpha = 0.05$ when there is no pre-registered directional hypothesis.

4. Example Statistical Result

For the illustrative counts above (100/1,000 versus 120/1,000), the pooled two-proportion z statistic is approximately 1.45 and the two-sided p-value is approximately 0.147. Since $p > 0.05$, this example does **not** provide sufficient evidence to reject H■ at the 5% significance level. The observed 20% relative lift may be commercially interesting, but the sample is not conclusive.

5. Interpretation & Decision

• Do not describe the variant as a proven winner based on this example alone.
• Check randomization, sample-ratio balance, tracking quality, and whether users were exposed to only one version.
• Estimate a confidence interval and evaluate whether the plausible lift is worth the implementation cost.
• If the result remains inconclusive, continue the test to a pre-planned sample size rather than stopping when the result looks favorable.

6. Presentation / Slide Deck Outline

Slide	Content
1 — Title	Business question, owner, date range, data source.

Slide	Content
2 — Context	Why conversion matters; current baseline and target.
3 — Data & method	Population, experiment setup, exclusions, KPI formula.
4 — Results	A/B table, conversion chart, absolute and relative lift.
5 — Statistical validation	Hypotheses, test, p-value, confidence interval, assumptions.
6 — Recommendation	Decision, limitations, next steps, owner and timeline.

7. Suggested Final Narrative

“The redesigned checkout showed a higher observed conversion rate in the sample, increasing from 10% to 12%. However, the two-sided test was not statistically significant at the 5% level ($p \approx 0.147$). We therefore recommend validating instrumentation and gathering the pre-planned sample before making a permanent rollout decision. The next analysis will break down results by device and payment method, while avoiding unplanned subgroup claims.”

Submission checklist

- ■ Presentation deck created using the actual project data
- ■ Hypothesis, test choice, p-value and confidence interval documented
- ■ Business recommendation includes limitations and next steps
- ■ 7–10 minute walkthrough recorded

Reminder: Replace all hypothetical figures with verified project results.